PHY223: Modern Physics Lab

Fall 2026

Instructor: Prof. Ryan Trainor

Office Hours: Mondays 1-2p, Thursdays 9-10a

Lab Goals:

- Build intuition and practical skills related to modern physics concepts such as scattering, light-matter interactions, sub-atomic particles, etc.
- Continue developing experimental skills for making precise measurements and testing variables
- Practice skills for data analysis including statistical tools, uncertainty analysis, and Python programming
- Develop communication skills for effective tabular, graphical, and narrative presentation of experimental results

Grading:

- Weekly Lab Activities — 9% of course grade
- Lab Report 1 — 2% of course grade
- Lab Report 2 — 4% of course grade
- **Total Lab:** — **15% of course grade**

Schedule:

Week	Date	Lab
1	Aug 26	Python & Inkscape Tutorials
2	Sep 2	BB Scattering Data Acquisition / Tutorials
3	Sep 9	BB Scattering Data Acquisition / Tutorials
4	Sep 16	BB Scattering Data Analysis / Effective Graphs
5	Sep 23	BB Scattering Data Analysis / Uncertainty
6	Sep 30	Writing a Report 1
7	Oct 7	Writing a Report 2
8	Oct 14	Lab 1
9	Oct 21	Lab 1
10	Oct 28	Lab 2
11	Nov 4	Lab 2
12	Nov 11	Writing a Report 3
13	Nov 18	Lab 3
		No Lab - Thanksgiving Break
14	Dec 2	Lab 3